

STATISTICAL ANALYSIS OF SALES DATA – DESCRIPTIVE & INFERENCEAL INSIGHTS

MINI PROJECT REPORT

Submitted By

Name: Aishwarya Jadhav

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ABSTRACT

This project analyzes sales data using descriptive and inferential statistical techniques. Measures such as mean, median, mode, variance, standard deviation, probability, confidence intervals, t-tests, and chi-square tests were applied to understand sales patterns and business performance. Various visualizations were also created to present the findings effectively.

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1. INTRODUCTION

Sales data analysis helps organizations evaluate performance, identify trends, and make informed business decisions. Statistical techniques provide meaningful insights from large volumes of data.

2. OBJECTIVES

- Analyze sales performance.
 - Calculate descriptive statistics.
 - Perform probability analysis.
 - Estimate population parameters using confidence intervals.
 - Conduct hypothesis testing.
 - Visualize sales trends and patterns.
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3. DATASET DESCRIPTION

Dataset Columns:

- Order_ID
- Product
- Category
- Quantity
- Unit_Price
- Region
- Month
- Total_Sales

Total_Sales was used as the primary variable for statistical analysis.

4. METHODOLOGY

Tools Used:

- Python
- Pandas
- NumPy
- SciPy

- Matplotlib

Statistical Methods:

- Mean
 - Median
 - Mode
 - Variance
 - Standard Deviation
 - Probability
 - Confidence Interval
 - One-Sample t-Test
 - Chi-Square Test
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5. DESCRIPTIVE STATISTICS

(Insert Output Screenshot)

Results:

Mean = 93745.26

Median = 91085.00

Mode = 1663.00

Variance = 6308413266.32

Standard Deviation = 79425.52

Interpretation:

The average sales value indicates overall business performance. Variance and standard deviation show the degree of variability in sales transactions.

6. PROBABILITY ANALYSIS

Probability of Total_Sales > ₹5000

Result = 0.9800

Interpretation:

This probability indicates how frequently high-value sales occur in the dataset.

7. CONFIDENCE INTERVAL

95% Confidence Interval:

Lower Limit = 71172.78

Upper Limit = 116317.74

Interpretation:

We are 95% confident that the true average sales lie within this range.

8. ONE-SAMPLE t-TEST

Hypotheses:

H_0 : Mean Sales = ₹5000

H_1 : Mean Sales $\neq$ ₹5000

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Interpretation:

If p-value < 0.05, the average sales significantly differ from ₹5000.

9. CHI-SQUARE TEST

Hypotheses:

H_0 : Region and Category are independent.

H_1 : Region and Category are associated.

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Interpretation:

The test determines whether sales region and product category are statistically related.

10. DATA VISUALIZATIONS

Histogram

(Insert Graph)

Boxplot

(Insert Graph)

Region-wise Sales Chart

(Insert Graph)

Category-wise Sales Pie Chart

(Insert Graph)

Monthly Sales Trend

(Insert Graph)

11. INTERPRETATION AND DISCUSSION

The analysis indicates the central tendency and spread of sales values. Statistical tests provide evidence regarding average sales performance and relationships among business variables. Visualizations reveal sales patterns across regions, categories, and months.

12. KEY FINDINGS

Chi-square test result was 5.8495

13. RECOMMENDATIONS

- Focus on high-performing regions.
- Improve marketing in low-performing regions.

- Increase inventory for high-demand categories.
 - Use statistical insights for future planning.
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14. CONCLUSION

The project successfully applied descriptive and inferential statistical methods to analyze sales data. The results provided meaningful insights into sales behavior and supported data-driven business decision-making.

15. REFERENCES

1. Python Documentation
2. Pandas Documentation
3. NumPy Documentation
4. SciPy Documentation
5. Matplotlib Documentation